

Ancillary proof: continuous-convolution rigidity through degree 72

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For $A \in \mathbb{C}[x]$, put

$$H_A(x) = \int_0^x A(t)A(x-t) dt = \sum_{i,j} a_i a_j \frac{i!j!}{(i+j+1)!} x^{i+j+1}, \quad A = \sum_i a_i x^i.$$

The integral is a polynomial identity. We prove the finite degree range used in the manuscript. The accompanying JSON supplies integer polynomial identities; the standalone replay uses exact arithmetic and checks every degree and every residual stratum. No assertion of rigidity in all degrees, or of simplicity of the discrete matching system, follows from this note.

Theorem 1. *Every nonzero complex polynomial of degree at most 72 satisfying $A \mid H_A$ is a monomial.*

1 The uniform prime-power reduction

Let ℓ be an odd prime, $e \geq 1$, $P = \ell^e$, and

$$2d+1 = P+g, \quad 0 \leq g < \ell, \quad g \text{ even.}$$

Write $A = x^d + a_1 x^{d-1} + \cdots + a_d$, with $a_0 = 1$. For $g > 0$, put $\delta = (g-1)/2$ and

$$S_{g,A}(x) = \sum_{r+s \leq g} \frac{g^{r+s}}{\delta^r \delta^s} a_r a_s x^{g-r-s}.$$

For $g = 0$, set $S_{0,A} = 1$.

Lemma 2. *The polynomial PH_A has coefficients integral at ℓ , and*

$$PH_A(x) \equiv u x^P S_{g,A}(x) \pmod{\ell}, \quad u = \frac{\overline{P(d!)^2}}{(P+g)!} \neq 0.$$

The assertion is an identity in the coefficient indeterminates of A .

Proof. For $\beta(i, j) = i!j!/(i+j+1)!$, Legendre's formula gives

$$v_\ell(\beta(i, j)^{-1}) = \sum_{h \geq 1} \left(\left\lfloor \frac{i+j+1}{\ell^h} \right\rfloor - \left\lfloor \frac{i}{\ell^h} \right\rfloor - \left\lfloor \frac{j}{\ell^h} \right\rfloor \right).$$

Each summand is zero or one. Since $i+j+1 \leq P+g < \ell P$, all summands with $h > e$ vanish. This proves integrality after multiplication by P . If $i+j+1 < P$, the summand at $h = e$ also vanishes, so the normalized coefficient reduces to zero.

For a remaining term, write $i = d - r$, $j = d - s$, $r + s \leq g$. For $1 \leq h \leq e$,

$$d \bmod \ell^h = \frac{\ell^h + g - 1}{2}.$$

Subtracting $r, s \leq g < \ell$ creates no borrow, and the sum of the two residues plus one is $\ell^h + g - r - s \geq \ell^h$. All e summands are therefore one. In particular the leading normalized coefficient is a unit. Dividing by that leading coefficient gives

$$\frac{(P + g)^{r+s}}{d^x d^s} \equiv \frac{g^{r+s}}{\delta^x \delta^s} \pmod{\ell}.$$

The denominators are units because their odd numerators have absolute value less than ℓ . This proves the coefficient formula. $\square$

Lemma 3. *Suppose that for each $1 \leq k \leq g$ no monic degree- k polynomial C , with $C(0) \neq 0$, satisfies*

$$C \mid S_{g,C} \quad \text{over } \overline{\mathbb{F}}_\ell.$$

Then rigidity holds in degree d . For $g = 0$, no residual hypothesis is needed.

Proof. A hypothetical nonmonomial solution gives an algebraic solution: the remainder equations have rational coefficients, and adjoining the inverse of a nonzero coefficient produces an affine system with a point over $\overline{\mathbb{Q}}$. At a place above ℓ , give the descending coefficient a_j weight j , and put $\eta = \min_{a_j \neq 0} v_\ell(a_j)/j$. After a finite extension choose t with $v_\ell(t) = -\eta$. Then $t^d A(x/t)$ is monic and integral, with a nonleading unit. This scaling preserves divisibility, since its convolution is $t^{2d+1} H_A(x/t)$.

Monic division of the integral normalized convolution has integral quotient. The preceding lemma therefore implies $\overline{A} \mid x^P S_{g,\overline{A}}$. If $g = 0$, this forces $\overline{A} = x^d$, contradicting the nonleading unit. Otherwise write $\overline{A} = x^{d-k} C$, $C(0) \neq 0$. The degree- g residual forces $1 \leq k \leq g$. Its descending coefficient list is the one used for C , so $C \mid S_{g,C}$, contrary to the hypothesis. The argument permits arbitrary ramification. $\square$

2 The finite certificates

For $g = 2, 4, 6$ and every $1 \leq k \leq g$, normalize $C(0) = 1$:

$$C = x^k + c_1 x^{k-1} + \cdots + c_{k-1} x + 1.$$

This loses no geometric point over the algebraic closure. Indeed $C_t(x) = t^k C(x/t)$ preserves the residual divisibility, and a suitable k -th root makes its constant coefficient one.

Take *all* coefficients of $\text{rem}_C S_{g,C}$, and clear their rational denominators, without discarding integer content. Call the resulting integer polynomials $F_{g,k,j}$. The file `certificate.json` contains integer polynomials $U_{g,k,j}$ and positive integers $D_{g,k}$ with

$$\sum_j U_{g,k,j} F_{g,k,j} = D_{g,k}.$$

When ℓ divides neither $D_{g,k}$ nor the original denominator scales, this identity rules out common zeros over every extension of $\mathbb{F}_\ell$. The JSON also contains the complete prime factorizations of these integers and recursive Lucas primality certificates.

For example,

$$S_{2,C} = x^2 + 8c_1x + 8c_1^2 - 16c_2.$$

In degree one the remainder is c_1^2 ; in degree two it is $7c_1x + 8c_1^2 - 17c_2$. The only possible exceptional odd primes greater than two in this criterion are 7 and 17. The same certificate procedure at gaps four and six supplies finite exceptional-prime covers. An exceptional prime is not declared a counterexample to characteristic-zero rigidity.

The stored coverage table gives, for every $1 \leq d \leq 72$, a triple (ℓ, e, g) , $g \in \{0, 2, 4, 6\}$, such that

$$2d + 1 = \ell^e + g, \quad \ell > g,$$

and, for $g > 0$, ℓ avoids every relevant certificate factor in every degree $1 \leq k \leq g$. For illustration, three choices filling gaps between the prime-index cases are

$$95 = 89 + 6, \quad 119 = 113 + 6, \quad 123 = 11^2 + 2.$$

The theorem follows from the full 72-row table and the preceding lemmas. Degree zero is immediate.

3 Independent replay

Run

`python verify_coverage.py`

in this directory, with SymPy installed. The program reconstructs the residual equations from the displayed falling-factorial kernel, checks every integer identity and denominator scale, verifies the prime factorizations and Lucas certificates, and validates every row of the coverage table. As an additional direct check, it rebuilds all 132348 ordered bilinear coefficients of PH_A for the 72 chosen degrees and compares them with the lemma. No computer algebra system beyond SymPy is required.

The final report explicitly records that the conclusion concerns $0 \leq \deg A \leq 72$, corresponding to ODE orders $2 \leq p \leq 73$ in the manuscript. It does not assert simplicity of those discrete matching schemes. The main paper's prime-index simplicity theorem and its separate order-five calculation supply their own arguments.